

Omar Kamel

AI / Machine Learning Engineer

📍 Monufia, Egypt

✉ omar13405@gmail.com

🌐 [linkedin.com/in/omarkamel](https://www.linkedin.com/in/omarkamel)

🐙 github.com/Omarkam3l

🌐 omarkamel-portfolio.vercel.app

PROFESSIONAL SUMMARY

AI/ML Engineer with strong experience building and deploying machine learning, deep learning, and NLP solutions. Skilled in end-to-end model development, data preprocessing, and MLOps orchestration. Proven track record delivering AI-driven applications using PyTorch, TensorFlow, Hugging Face, LangChain, and cloud platforms. Looking to bring scalable AI expertise to an impactful role.

TECHNICAL SKILLS

Languages

Python, SQL, Bash

Frameworks

PyTorch, TensorFlow, Keras, Scikit-learn

AI Tools

Hugging Face Transformers, OpenAI API, LangChain, RAG

Big Data & Cloud

Spark (PySpark), AWS

MLOps & Deployment

Docker, FastAPI, MLflow, Git

Data Science

Feature engineering, model tuning, evaluation, visualization (Matplotlib, Seaborn, Plotly)

SELECTED PROJECTS

1. AI-Powered Food Waste Reduction Platform (Kathir)

- Built a multi-agent AI platform using LangGraph and LangChain to connect restaurants, users, and NGOs for surplus meal distribution.
- Implemented hybrid semantic + keyword search using ChromaDB embeddings and RRF for accurate meal retrieval.
- Developed personalized recommendations based on user preferences, history, dietary restrictions, budget, and location.
- Integrated FastAPI with Supabase (PostgreSQL) for authentication, real-time data, and scalable backend services.

Tech Stack: Flutter, FastAPI, LangGraph, LangChain, ChromaDB, Supabase, PostgreSQL, Hugging Face, PyTorch

🐙 GitHub Repository: github.com/Omarkam3l/Kathir_final

🔗 Live Demo (Chatbot): huggingface.co/spaces/omark3405/Kathir_V2

2. RAG Chatbot

- Developed a Retrieval-Augmented Generation (RAG) chatbot supporting PDF, DOCX, CSV, and URL ingestion.
- Built vector search pipeline using FAISS and integrated generative models for context-aware responses.
- Created a Streamlit interface enabling real-time interaction with custom document datasets.

Tech Stack: Hugging Face, Pinecone, FAISS, PyTorch, Streamlit

🐙 GitHub Repository: github.com/Omarkam3l/rag-chatbot

3. Big Data Analytics Pipeline

- Built a full-fledged big data pipeline processing streaming flight data using Kafka, Spark, and Hive.
- Performed data ingestion, real-time analytics, data warehousing, and visualization in Python/Databricks.

Tech Stack: Apache Spark (PySpark), Apache Kafka, Hive, Python, Databricks

🐙 GitHub Repository: github.com/Omarkam3l/big-data-pipeline

5. AdventureWorks2022 ETL & BI Pipeline

- Designed and implemented an end-to-end ETL pipeline using SSIS to extract, transform, and load data.
- Built optimized star-schema data models and developed Power BI dashboards for sales and performance insights.

Tech Stack: SSIS, SQL Server, Power BI, T-SQL

🐙 GitHub Repository: github.com/Omarkam3l/adventureworks-etl-bi

PROFESSIONAL EXPERIENCE

AI & Data Science Trainee – Digital Egypt Pioneers Program (DEPI)

April 2024 – October 2024 | Ministry of Communications and Information Technology (MCIT), Egypt

- Completed a government-sponsored training program specializing in AI, Data Science, and Microsoft Data Engineering.
- Developed and deployed data pipelines for extraction, transformation, and visualization using Python and SQL.
- Worked on case studies integrating machine learning models and business intelligence dashboards.

Tech Stack: Python, SQL, Power BI, Azure, Machine Learning, Data Engineering

EDUCATION

BSc in Computers & AI Engineering – Monufia National University

Currently Fourth Year (Graduation expected 2026)

CERTIFICATIONS

- Digital Egypt Pioneers Program (DEPI) — AI & Data Science: Microsoft Data Engineer Job Profile
- Egyptian Talent Academy (ETA) — Artificial Intelligence (AI) Track
- Udemy — Machine Learning & Data Science Diploma
- DeepLearning.AI TensorFlow Developer